Dr. Bansilal Bairwa

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85	510	15	11	12	8
SCOPUS DOCUMENTS	SCOPUS CITATIONS	H-INDEX	PATENTS FILED	KSCST PROJECTS	AWARDS

Scopus metrics computed from the Scopus export dated 05 October 2026. Google Scholar: 608 citations, h-index 15, i10-index 23.

PROFESSIONAL SUMMARY

Assistant Professor in the School of Electrical and Electronics Engineering, REVA University, Bengaluru (since December 2020), with a Ph.D. from MNIT Jaipur on lithium-ion battery modelling for electric-vehicle drive cycles. My research spans **electric-vehicle battery modelling, battery management systems (BMS), AI/ML-based state-of-charge and state-of-health estimation, battery thermal management, EV powertrain modelling** and **robotics**. I have authored **85 Scopus-indexed documents** (510 citations, h-index 15), including a first-author article in *Scientific Reports* (Nature Portfolio), filed **8 patents**, and led **12 KSCST-funded student research projects** (10 as PI). My work has been recognised with **two Best IEEE Paper Awards**, three Best Poster Awards (including IIT Kanpur and CoreEL Technologies) and the REVA University **Best Researcher Award (2024)**. A MATLAB/Simulink specialist, I have delivered **26 invited sessions** as resource person and keynote speaker in AICTE ATAL FDPs, institutional FDPs and workshops, and serve the institution as GATE Coordinator, KSCST Project Coordinator and industry-collaboration coordinator (Bosch Global Software Technologies, MathWorks, CoreEL Technologies).

VIRTUAL LABORATORY DEVELOPMENT

- **EV MASTER LAB**: Interactive Electric Vehicle System Engineering Virtual Laboratory (battery modelling, motor control, regenerative braking, power-flow & real-time simulation). https://aya430.github.io/EV-MASTER-LAB/#/home
- **Robotics & Instrumentation Virtual Lab**: Interactive virtual lab for robotics and instrumentation concepts. https://aya430.github.io/robotics-instrumentation-virtual-lab/index.html

RESEARCH INTERESTS

- Lithium-ion battery modelling (equivalent-circuit, thermal, electro-thermal)
- Battery management systems; SoC / SoH estimation
- Deep learning for battery health diagnostics (MLP, GRU, TCN, LSTM)
- Battery thermal management (air and PCM-based cooling)
- EV powertrain modelling, motor drives and drive-cycle-based sizing
- Robotics: kinematics, dynamics and control (6-DoF, humanoid)
- Fuel-cell systems and renewable-energy integration
- Embedded and IoT systems; hardware integration with MATLAB

ACADEMIC QUALIFICATIONS

2026 **PG Certification in Robotics** — Palakkad, Kerala
Indian Institute of Technology Palakkad
Advisor: Prof. Santhakumar Mohan

2021 **Ph.D.** — Jaipur, Rajasthan
Centre for Energy and Environment, Malaviya National Institute of Technology (MNIT) Jaipur
Thesis: *Modeling and Performance Analysis of Li-ion Battery for Electric Vehicle (EV) Drive Cycles*
Supervisor: Dr. Kapil Pareek | CGPA: 8.66

2016 **M.Tech. in Power and Energy Systems** Surathkal, Karnataka
National Institute of Technology Karnataka (NITK) Surathkal
Thesis: *Comparative Analysis of Control Strategies for Dynamic Voltage Restorer (DVR)*
Supervisor: Dr. A. Karthikeyan | CGPA: 6.05 | GATE Scholarship (AICTE, Govt. of India)

2014 **B.Tech. in Electrical Engineering** Chittorgarh, Rajasthan
Mewar University
CGPA: 7.43

2010 | 2008 **Senior Secondary (XII, PCM): 69.85% | Secondary (X): 62.00%** Rajasthan
Board of Secondary Education, Rajasthan

PROFESSIONAL AND RESEARCH EXPERIENCE

Dec 2020 – Present **Assistant Professor** Bengaluru, Karnataka
School of Electrical and Electronics Engineering, REVA University

- Teaching UG and PG courses in robotics, electric vehicles, energy storage, machine learning and Python; in-charge of the MATLAB & Simulink, Python Programming and Signal Processing laboratories.
- As KSCST Project Coordinator, facilitated 35 sanctioned student projects (2021–2025); the School recorded the highest number of KSCST-funded projects in Karnataka in 2023–24 and 2024–25.
- Member of the *AI First University* team, REVA University.

2017 – 2019 **Junior Research Fellow (SERB, Govt. of India)** Jaipur, Rajasthan
Centre for Energy and Environment, MNIT Jaipur

Project: *Development of Flexible Supercapacitor for Flexible & Wearable Electronics* (PI: Dr. Kapil Pareek).
Also worked as Statistical Analyst in the Solar Photovoltaic Division (PI: Prof. Jyotirmay Mathur), Centre for Energy and Environment, MNIT Jaipur.

HONOURS AND AWARDS

- **Best Poster Award** – "Integrating Hardware with MATLAB", CoreEL Technologies, **2025**.
- **Best Poster Award** – "Design and Development of Cooling System for Electric Vehicle Battery Bank", Indian Institute of Technology Kanpur, **2024**.
- **Best Researcher Award** (Teacher's Day 2024) – conferred by the Hon'ble Chancellor, REVA University, Bengaluru, **2024**.
- **Best IEEE Paper Award** – "Modeling and Simulation of Electric Vehicle Powertrain for Dynamic Performance Evaluation", IEEE ICDCECE-2023, Ballari Institute of Technology and Management, **2023**.
- **Best IEEE Paper Award** – "Electrical Equivalent Circuit Modeling of Lithium-Ion Battery", IEEE ICSTCEE, REVA University, Bengaluru, **2022**.
- **Best Poster Award** – "Electrical Equivalent Circuit Modeling of Lithium-Ion Battery for Electric Vehicle", Industry Academia Conclave, MNIT Jaipur, **2019**.
- **Junior Research Fellowship**, Science and Engineering Research Board (SERB), Govt. of India, MNIT Jaipur, **2017**.
- **GATE Scholarship**, AICTE, Govt. of India, for M.Tech. at NIT Surathkal, **2014**.

KSCST FUNDED PROJECTS

12 projects funded by the Karnataka State Council for Science and Technology (KSCST) – 10 as Principal Investigator and 2 as Co-PI; total sanctioned amount **INR 70,000**.

Year	Project Title (Project Code)	Role	INR
2026	Advanced Battery Management System with Digital Twin for Fault Detection and Accurate SOC Estimation (49S_MTECH_0007)	PI	6,500
2026	Real-Time Monitoring and Thermal Management of Lithium-ion Battery Bank with Hybrid Air-PCM Cooling System (49S_MTECH_0003)	PI	6,500
2026	Real-Time Distance-Adaptive Smart Trolley for Continuous Human Tracking (49S_BE_0871)	PI	5,000
2026	Deep Learning-Powered Visual Aid System for Blind Navigation (49S_BE_1384)	PI	5,000
2026	Design and Implementation of an Embedded Control Platform for Air and PCM-Based BESS Cooling (49S_BE_1872)	PI	5,000

Year	Project Title (Project Code)	Role	INR
2026	Fault Detection in EV Motors Using AI (49S_BE_2828)	PI	5,000
2025	Development of Automatic Solar Panel Cleaning System	PI	4,500
2025	Multifunctional Sign Hand Gloves for Women's Safety	Co-PI	4,500
2024	Thermal Management System for Electric Vehicle Battery Bank: Improving Performance and Safety	PI	5,500
2024	Unmanned Smart Fire Fighting System for Forest and Wildlife Protection	Co-PI	5,500
2023	Development of Low-Cost Modified Hybrid Bike for Rural Area	PI	7,000
2022	Adaptable Skating Electric Vehicle for Physically Challenged People	PI	10,000

PATENTS (8 INDIAN PATENT APPLICATIONS)

No.	Title	Year	Application No.
P1	Low-Cost Solar Powered Battery Charger	2023	202341025713
P2	Development of Smart Automated Fire Fighting Robot	2023	202241077491
P3	Designing a Simulation for Cooler Section in Cement Industry Using PLC and VFD	2023	202241077487
P4	Petro-Hybrid Electric Vehicle	2022	202241077504
P5	Automatic Mask and Sanitizer Vending Machine	2022	202241055595
P6	Voice and Hand Gesture Controlled Robotic Vehicle	2021	202141033053
P7	Battery Swapping Mobile Service Station for Electric Vehicles Application	2021	202141031986
P8	Modeling and Simulation of Wireless Electric Vehicle Charging Using Solar and Wind Energy	2021	202141028751

PUBLICATIONS

Summary (Scopus, 05 October 2026): 85 documents – 4 journal articles, 79 conference papers, 1 book chapter and 1 editorial; 498 citations; h-index 15. Additionally, 6 book chapters with River Publishers (2021).

Journal Articles

[J1] **B. Bairwa**, K. Pareek, and V. K. Jadoun, "Cycle based state of health estimation of lithium ion cells using deep learning architectures," *Scientific Reports* (Nature Portfolio), vol. 15, no. 1, 2025. doi:10.1038/s41598-025-20995-7

[J2] B. S. Sagar, S. Hampannavar, D. B, and **B. Bairwa**, "Applications of battery management system (BMS) in sustainable transportation: A comprehensive approach from battery modeling to battery integration to the power grid," *World Electric Vehicle Journal*, vol. 13, no. 5, art. 80, 2022. doi:10.3390/wevj13050080

[J3] **B. Bairwa**, K. Pareek, and S. Hampannavar, "Investigation on lithium ion battery equivalent circuit models for dynamic load profiles," *Energy Storage* (Wiley), vol. 3, no. 2, 2021. doi:10.1002/est2.231

Book Chapters

[B1] R. Manojkumar, P. E. Gundgurti, T. A. S. Srinivas, R. B. Pittala, A. S. Konda, and **B. Bairwa**, "Building ethical awareness in AI-enhanced educator training for responsible and inclusive learning environments," in *Ethical Leadership and Pedagogical Transformation in AI-Driven Education*, pp. 139–171, 2026. doi:10.4018/979-8-2600-1840-8.ch005

[B2] A. Banik, **B. Bairwa**, S. Saahithi, and H. Rajini, "Application of solar energy for wireless power transfer," River Publishers, 2021.

[B3] **B. Bairwa**, S. Kumar, and V. S. Kaushik, "Assessment of 50 kW grid connected solar power plant with experimental validation," River Publishers, 2021.

[B4] **B. Bairwa**, S. Kumar, S. Hampannavar, and C. N. Raghu, "Modeling of PWM technique for three phase voltage source inverter," River Publishers, 2021.

[B5] S. Rao, **B. Bairwa**, D. Ahmed, and S. Staines, "Regenerative braking system using a DC/DC buck-boost converter," River Publishers, 2021.

[B6] K. Nethra, **B. Bairwa**, and C. S. V. Himabindu, "Investigation on inverter arc welding circuit," River Publishers, 2021.

[B7] A. Banik, **B. Bairwa**, and N. Mamatha, "A literature review on PV inverter topologies connected to grid," River Publishers, 2021.

Editorial

[E1] V. Subburaj, R. Dash, K. Jyotheeswara Reddy, and **B. Bairwa**, "Preface: International Conference on Recent Trends in Electrical, Electronics & Computer Engineering for Environmental and Sustainable Development (ICRTEEC-2021)," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/12.0010036

Conference Papers (Scopus-indexed: 79)

2026

[C1] **B. Bairwa**, J. P. Sridhar, S. Thingbaijam, and V. K. Jadoun, "Thermal model of a 18650 cell for SoC prediction using HPPC and CCCV," *3rd Int. Conf. on Integrated Intelligence and Communication Systems (ICIICS)*, 2026. doi:10.1109/ICIICS67880.2026.11483573

[C2] **B. Bairwa**, K. M. Vaishnavi, P. Sachin, K. Sapthami, S. Reigo Melwin, and V. K. Jadoun, "Balancing latent capacity and heat generation in PCM-cooled Li-ion 5S6P modules," *Int. Conf. on Recent Advancement in Electrical, Computer and Communication Technologies (IECCT)*, 2026. doi:10.1109/IECCT68664.2026.11541745

[C3] G. C. Tanish Prajwal, **B. Bairwa**, P. S. Tejas, S. S. Joshi, A. Reddy, and S. Ashok, "Mathematical modeling and kinematic analysis of a 20-DOF humanoid robot," *IECCT*, 2026. doi:10.1109/IECCT68664.2026.11541577

2025

[C4] **B. Bairwa**, K. Pareek, J. P. Sridhar, and U. R. Yaragatti, "Design and development of cooling system for electric vehicle batteries," *5th IEEE Int. Conf. on Sustainable Energy and Future Electric Transportation (SeFeT)*, 2025. doi:10.1109/SEFET65155.2025.11255175

[C5] **B. Bairwa**, Sahebgouda, S. Byelle, S. Faiz, and S. Ahmed, "Modeling and performance analysis of a fuel cell system under pressure variations," *5th IEEE SeFeT*, 2025. doi:10.1109/SEFET65155.2025.11255021

[C6] **B. Bairwa**, C. B. Chandrika, A. H. Monisha, N. B. Gowda, and B. Ullas, "Predictive system for early failure detection in EV battery bank cells," *Int. Conf. on Advances in Renewable Energy and Electric Vehicles (AREEV)*, pp. 32–37, 2025. doi:10.1109/AREEV64307.2025.10925196

[C7] J. P. Sridhar, P. Pramila, K. Nagarathna, S. Ashok Kumar, and **B. Bairwa**, "Electro thermal modeling of runaway propagation and barrier efficacy in multi cell modules," *Int. Conf. on Electrical, Electronics, and Computer Science with Advance Power Technologies (ICE2CPT)*, 2025. doi:10.1109/ICE2CPT66440.2025.11340107

[C8] S. N. Devi, **B. Bairwa**, S. Thingbaijam, and J. P. Sridhar, "Drive cycle-based system-level sizing of electric vehicle components for Indian drive cycles," *IEEE Int. Conf. on Vehicular Technology and Transportation System (ICVTTS)*, 2025. doi:10.1109/ICVTTS67119.2025.11296414

[C9] Praveenkumar, **B. Bairwa**, Krishna, and Veena, "Single to quad-motor architectures in battery electric vehicle using MATLAB/Simulink," *IEEE ICVTTS*, 2025. doi:10.1109/ICVTTS67119.2025.11296472

[C10] S. Thingbaijam, **B. Bairwa**, N. Sougaijam, and J. P. Sridhar, "Torque-speed behavior analysis of five electric motors using MATLAB Simulink," *Int. Conf. on Electrical Energy Systems (ICEES)*, pp. 343–348, 2025. doi:10.1109/ICEES67011.2025.11213366

[C11] N. Himabindu, J. P. Sridhar, S. Hemanth Kumar, C. A. Swathi, and **B. Bairwa**, "Kinematic and dynamic control of a 6-DoF robot using MATLAB and Python," *IEEE ICVTTS*, 2025. doi:10.1109/ICVTTS67119.2025.11296509

[C12] **B. Bairwa**, A. Patil, K. Keerthan, S. Byelle, and Eknath, "PID-based distance-adaptive control of a smart trolley for continuous human tracking," *IEEE ICVTTS*, 2025. doi:10.1109/ICVTTS67119.2025.11296417

[C13] **B. Bairwa**, B. Reddy, A. Janali, Ambrish, and M. Chandana, "Deep learning-powered visual aid system for blind navigation," *ICE2CPT*, 2025. doi:10.1109/ICE2CPT66440.2025.11340459

[C14] K. Nethra, **B. Bairwa**, N. Himabindu, and V. K. Jadoun, "Modeling and analysis of a proton exchange membrane fuel cell stack," *7th Int. Conf. on Energy, Power and Environment (ICEPE)*, 2025. doi:10.1109/ICEPE65965.2025.11139455

[C15] **B. Bairwa**, R. N. Gautham, N. Himabindu, and T. A. Kushal, "Real-time control of servomotor using Arduino and MATLAB," *AREEV*, pp. 65–70, 2025. doi:10.1109/AREEV64307.2025.10925201

[C16] **B. Bairwa**, S. R. Biradar, S. Ahmed, S. Faiz, and R. Audithya Iyer, "Voice-activated multi-floor elevator with proximity-based entry for touchless navigation," *ICEES*, pp. 337–342, 2025. doi:10.1109/ICEES67011.2025.11213000

[C17] **B. Bairwa**, F. Pathan, Noorsaba, and V. K. Jadoun, "Real-time smart home monitoring and control using Raspberry Pi, sensors, and edge AI," *ICEES*, pp. 349–354, 2025. doi:10.1109/ICEES67011.2025.11212909

[C18] **B. Bairwa**, S. Magadum, T. C. Salian, V. M. Bagali, Y. Sandeep, and M. G. Sekhar, "IoT-enabled glove for women's safety and sign language translation," *Int. Conf. on Intelligent Computing and Knowledge Extraction (ICICKE)*, 2025. doi:10.1109/ICICKE65317.2025.11136270

[C19] S. P. Jadhav, V. N. Kalkhambkar, **B. Bairwa**, D. Kanase, R. Bulbule, and S. D. Chavan, "Arduino-based dual-axis solar tracker with weather condition monitoring to enhance agricultural energy efficiency," *Global Conf. in Emerging Technology (GINOTECH)*, 2025. doi:10.1109/GINOTECH63460.2025.11076835

[C20] R. Nagar, S. N. Suresha, and **B. Bairwa**, "Automated parking lot management system using number plate recognition," *AREEV*, pp. 112–117, 2025. doi:10.1109/AREEV64307.2025.10925183

2024

[C21] **B. Bairwa**, K. Jampala, J. P. Sridhar, and D. Silas Stephen, "Temperature dependent investigation on two RC equivalent circuit battery model," *15th Int. Conf. on Computing Communication and Networking Technologies (ICCCNT)*, 2024. doi:10.1109/ICCCNT61001.2024.10725261

[C22] **B. Bairwa**, S. A. Saleem, and K. Jampala, "Machine learning for battery charging: From prediction to categorization," *ICCCNT*, 2024. doi:10.1109/ICCCNT61001.2024.10725441

[C23] **B. Bairwa** and R. Soujanya, "Mathematical modeling and analysis of electric vehicle battery," *1st Int. Conf. on Innovative Sustainable Technologies for Energy, Mechatronics and Smart Systems (ISTEMS)*, 2024. doi:10.1109/ISTEMS60181.2024.10560087

[C24] **B. Bairwa**, P. J. Harshith, B. M. Vikas, and H. T. Likhith, "Energy efficiency assessment of slow and fast charging pattern in battery system," *1st Int. Conf. on Sustainability and Technological Advancements in Engineering Domain (SUSTAINED)*, pp. 950–956, 2024. doi:10.1109/SUSTAINED63638.2024.11073863

[C25] G. N. Abhishek, **B. Bairwa**, D. S. Akash, M. P. Kumar, M. K. Sai Pradeep, and G. Prashanth, "Optimizing electric vehicle performance using chain gear ratio approach," *ISTEMS*, 2024. doi:10.1109/ISTEMS60181.2024.10560175

[C26] **B. Bairwa**, Chandrashekhar, M. Sai Diksheeth, S. B. Shivayya, and C. M. Varun, "Design and development of self-recharging electric bicycle," *ICCCNT*, 2024. doi:10.1109/ICCCNT61001.2024.10724801

[C27] S. Byelle, **B. Bairwa**, C. N. Raghu, R. Santosh Chakravarty, V. H. Sank, and U. K. Chaturya, "Modeling and performance analysis of a 350 kW solar power plant," *ICCCNT*, 2024. doi:10.1109/ICCCNT61001.2024.10725710

[C28] **B. Bairwa**, V. Kumar, Z. Omar, A. Sayed, and V. Kumar, "Tomato plant disease detection using image processing for agriculture application," *IEEE Int. Conf. on Big Data and Machine Learning (ICBDML)*, pp. 270–276, 2024. doi:10.1109/ICBDML60909.2024.10577337

[C29] N. Himabindu, S. S. Vishnu, M. Vishveswaran, R. Swaroop, and **B. Bairwa**, "Design and implementation of RFID-driven parking management systems," *IEEE ICBDML*, pp. 97–102, 2024. doi:10.1109/ICBDML60909.2024.10577297

[C30] D. B. Anil Kumar, **B. Bairwa**, H. Anusha, M. R. Aravind, S. Kumar, and N. Doddabasappa, "Investigation on smart agriculture monitoring system using IoT," *ICCCNT*, 2024. doi:10.1109/ICCCNT61001.2024.10724229

[C31] S. Dwivedi, **B. Bairwa**, and P. Shriharinayaka, "Design and development of IoT enabled health monitoring system using Node MCU," *4th Int. Conf. on Intelligent Technologies (CONIT)*, 2024. doi:10.1109/CONIT61985.2024.10626290

2023

[C32] **B. Bairwa**, B. Supraja, C. Pratiksha, and A. Sumanth, "Modeling and simulation of electric vehicle powertrain for dynamic performance evaluation," *2nd IEEE Int. Conf. on Distributed Computing and Electrical Circuits and Electronics (ICDCECE)*, 2023. doi:10.1109/ICDCECE57866.2023.10150956 **(Best Paper Award)**

[C33] **B. Bairwa**, P. ShriHarinayaka, B. S. Sagar, and P. Ashwini Kumari, "Temperature dependent capacity fade prediction of electric vehicles batteries," *IEEE Int. Conf. on Recent Trends in Electronics and Communication (ICRTEC)*, 2023. doi:10.1109/ICRTEC56977.2023.10111909

[C34] **B. Bairwa**, K. A. Manohar, M. M. Magadum, S. Keerthi, and R. Chitrashree, "Modeling of low cost battery charge controller for stationary to mobile applications," *IEEE ICRTEC*, 2023. doi:10.1109/ICRTEC56977.2023.10111855

[C35] H. Rajini, **B. Bairwa**, A. Banik, and S. Jagadeesh, "Development of on road charging system for electric vehicle applications," *IEEE ICRTEC*, 2023. doi:10.1109/ICRTEC56977.2023.10111900

[C36] H. Rajini, **B. Bairwa**, A. Banik, and K. A. Manohar, "Modeling and simulation of inverters for electric vehicle application," *Int. Conf. for Advancement in Technology (ICONAT)*, 2023. doi:10.1109/ICONAT57137.2023.10080144

[C37] **B. Bairwa**, M. Murari, M. Shahapur, M. R. Kavya, and M. F. Khan, "Speed control of BLDC motor using PI controller," *ICONAT*, 2023. doi:10.1109/ICONAT57137.2023.10080074

[C38] **B. Bairwa**, M. Murari, M. Sahapur, M. R. Kavya, and M. F. Khan, "Drive cycle based speed control of BLDC motor using pulse width modulation," *IEEE ICRTEC*, 2023. doi:10.1109/ICRTEC56977.2023.10111848

[C39] R. Soujanya, **B. Bairwa**, B. S. Sagar, S. Sharma, and A. R. Patil, "Efficient power generation in electric vehicles: A chain gear ratio approach," *IEEE 2nd Int. Conf. on Industrial Electronics: Developments and Applications (ICIDeA)*, pp. 7–12, 2023. doi:10.1109/ICIDeA59866.2023.10295259

[C40] K. Nethra, D. V. Kumaar, G. Ajith, B. N. Reddy, **B. Bairwa**, and N. Himabindu, "Development of online monitoring system for wireless charging of electric vehicles," *4th IEEE Int. Conf. on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, 2023. doi:10.1109/ICSTCEE60504.2023.10585138

[C41] N. Himabindu, K. Nethra, **B. Bairwa**, and S. Potturi, "Enhancing battery life for processing ECG in cardiac pacemaker," *IEEE Renewable Energy and Sustainable E-Mobility Conf. (RESEM)*, 2023. doi:10.1109/RESEM57584.2023.10236382

[C42] S. Magadum, S. S. Reddy, **B. Bairwa**, and A. Sayeed, "Optimizing power system stability in multi-machine systems: A heuristic approach to power system stabilizer design," *2nd IEEE ICDCECE*, 2023. doi:10.1109/ICDCECE57866.2023.10150900

[C43] C. N. Raghu, **B. Bairwa**, G. Raghavendra, and K. A. Manohar, "Mathematical modeling and investigation on faults in power system," *ICONAT*, 2023. doi:10.1109/ICONAT57137.2023.10080232

[C44] B. S. Sagar, **B. Bairwa**, D. R. Sahana, N. B. Sujan, S. K. Sadari, and E. Sukrutha, "Optimization of solar-wind hybrid energy system with IoT-based source selector for enhanced power generation in smart homes," *4th IEEE Global Conf. for Advancement in Technology (GCAT)*, 2023. doi:10.1109/GCAT59970.2023.10353274

[C45] C. Jaya Krishna, S. Jaheer Abbas, **B. Bairwa**, V. MohithReddy, D. Raghavendra, and J. D. Dhanush, "Design and development of low-cost solar dryer for agriculture applications," *IEEE RESEM*, 2023. doi:10.1109/RESEM57584.2023.10236056

[C46] D. B. Anil Kumar, N. Doddabasappa, **B. Bairwa**, C. S. Anil Kumar, G. Raju, and Madhu, "IoT-based water harvesting, moisture monitoring, and crop monitoring system for precision agriculture," *2nd IEEE ICDCECE*, 2023. doi:10.1109/ICDCECE57866.

2023.10150893

[C47] B. S. Sagar, **B. Bairwa**, S. K. Hulsure, S. S. Walikhindi, M. Supritha, and S. S. Patgar, "Optimizing pesticide spraying efficiency through disease leaf detection mechanism," *IEEE ICIDeA*, pp. 19–25, 2023. doi:10.1109/ICIDeA59866.2023.10295256

[C48] K. Sohan, V. K. Angadi, M. Vaishak, and **B. Bairwa**, "Development of robot for intelligent surrounding recognition and direction control," *IEEE Int. Conf. on Advances in Electronics, Communication, Computing and Intelligent Information Systems (ICAECIS)*, pp. 676–681, 2023. doi:10.1109/ICAECIS58353.2023.10170237

[C49] **B. Bairwa**, N. Vershini, V. K. Angadi, and M. Vijay Kumar, "Development of real time monitoring and tracking system for vehicles application," *ICONAT*, 2023. doi:10.1109/ICONAT57137.2023.10080069

[C50] Y. B. Shankar, K. Baiju, and **B. Bairwa**, "IoT enabled innovative accident detection and rescue system," *IEEE ICAECIS*, pp. 663–667, 2023. doi:10.1109/ICAECIS58353.2023.10170604

[C51] S. Rohan, **B. Bairwa**, B. G. Gnandeep, and S. Gaganashree, "IoT based aid and health monitoring for physically disabled person," *Int. Conf. on Computational Intelligence and Sustainable Engineering Solution (CISES)*, pp. 306–311, 2023. doi:10.1109/CISES58720.2023.10183516

[C52] B. M. Praveen, B. S. Venkatesh Murthy, U. Rajesh Bhat, M. Mallikarjuna, R. S. Anandraj, and **B. Bairwa**, "Real time health monitoring system using IoT," *IEEE RESEM*, 2023. doi:10.1109/RESEM57584.2023.10236214

[C53] S. Rohan, R. Mohan Yadav, S. Siddharth, and **B. Bairwa**, "Smart traffic fines management system using GSM module," *IEEE RESEM*, 2023. doi:10.1109/RESEM57584.2023.10236369

[C54] B. S. Venkateshmurthy, M. Palati, P. Shriharinayak, and **B. Bairwa**, "Design and development of smart locker for domestic applications," *IEEE ICIDeA*, pp. 1–6, 2023. doi:10.1109/ICIDeA59866.2023.10295207

[C55] K. Nethra, **B. Bairwa**, N. Himabindu, and P. Sudharanu, "The evolution of street lighting: From manual to self-activation via object movement," *4th IEEE ICSTCEE*, 2023. doi:10.1109/ICSTCEE60504.2023.10585246

[C56] D. B. Anil Kumar, **B. Bairwa**, B. Vamsidath, V. Ujwal, P. Tharun, and T. Rao, "HDL programming and sequential circuitry for multi-core RISC-V processor," *Int. Conf. on Smart Systems for Applications in Electrical Sciences (ICSSES)*, 2023. doi:10.1109/ICSSES58299.2023.10201166

2022

[C57] **B. Bairwa**, K. Pareek, K. A. Manohar, and E. Nanditha Varshini, "Electrical equivalent circuit modeling of lithium-ion battery," *3rd Int. Conf. on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, 2022. doi:10.1109/ICSTCEE56972.2022.10100238 **(Best Paper Award)**

[C58] **B. Bairwa**, K. Pareek, M. Sarvagya, and U. R. Yaragatti, "Analysis of leakage current mechanism in supercapacitor with experimental approach," *4th Int. Conf. on Advances in Electronics, Computers and Communications (ICAECC)*, 2022. doi:10.1109/ICAECC54045.2022.9716664

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[C60] **B. Bairwa**, C. N. Raghu, P. Ashwini Kumari, and S. Oommen, "Comparative analysis of control strategies for dynamic voltage restorer," *Int. Conf. on Trends in Electrical, Electronics, Computer Engineering (TEECCON)*, pp. 166–171, 2022. doi:10.1109/TEECCON54414.2022.9854834

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[C62] B. S. Sagar, **B. Bairwa**, K. K. Reddy, V. Jyothi, R. Keerthana, and R. Manasa, "Design of IoT based battery monitoring system," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/5.0094107

[C63] K. Nethra, **B. Bairwa**, G. S. Sai Kiran, K. B. Tejas Gowda, K. M. Tharun, and V. Yeshwanth, "Modeling and simulation of wireless electric vehicle charging using solar and wind energy," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/5.0099858

[C64] Ahmed, S. Swaraj, **B. Bairwa**, and C. N. Raghu, "Hub motor powered petro-electric hybrid vehicle," *IEEE 2nd Mysore Sub Section Int. Conf. (MysuruCon)*, 2022. doi:10.1109/MysuruCon55714.2022.9972549

[C65] B. S. Sagar, **B. Bairwa**, Ganavi M, N. Arshiya, B. Sharavani, and Jayashalini, "Microcontroller based automatic dual axis solar tracking and cleaning system," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/5.0094108

[C66] K. A. Manohar, **B. Bairwa**, U. R. Yaragatti, and C. N. Raghu, "PV power prediction and investigation for movable solar PV module," *IEEE MysuruCon*, 2022. doi:10.1109/MysuruCon55714.2022.9972462

[C67] S. Kumar, **B. Bairwa**, S. Hampannavar, and B. Ankaiah, "Power quality issues by nonlinear loads and use of K-rated transformer," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/5.0094206

[C68] A. P. Kumari, P. Kumar S, K. R. Deepa, P. Geethanjali, **B. Bairwa**, and H. Ravishankar, "Modelling and performance analysis of photovoltaic system for low cost lawn cutting application," *IEEE MysuruCon*, 2022. doi:10.1109/MysuruCon55714.2022.9972438

[C69] A. Burri, C. S. Honnesh, G. Sachin, M. H. Ananda, and **B. Bairwa**, "Design and simulation of high speed electronic circuit breaker," *AIP Conference Proceedings*, vol. 2461, 2022. doi:10.1063/5.0093453

[C70] **B. Bairwa**, S. Rathod, U. R. Yaragatti, and K. A. Manohar, "Development of fault detection method in cable using Arduino UNO," *7th IEEE Int. Conf. on Recent Advances and Innovations in Engineering (ICRAIE)*, pp. 472–476, 2022. doi:10.1109/ICRAIE56454.2022.10054271

[C71] **B. Bairwa**, K. Y. Reddy, D. Deva, K. Kavya, and K. A. Manohar, "Development of IoT based GSM controller for agriculture applications," *4th Int. Conf. on Emerging Research in Electronics, Computer Science and Technology (ICERECT)*, 2022. doi:10.1109/ICERECT56837.2022.10060357

[C72] S. Potturi, **B. Bairwa**, K. Mallikarjuna, and C. Mayur, "Development of solar coupled synchronous buck converter," *ICERECT*, 2022. doi:10.1109/ICERECT56837.2022.10060197

[C73] K. M. Bhargavi, **B. Bairwa**, and S. Saahithi, "Extended Kalman filter-based EV charging station with PV, wind, and battery integration," *IEEE North Karnataka Subsection Flagship Int. Conf. (NKCon)*, 2022. doi:10.1109/NKCon56289.2022.10126754

[C74] A. Banik, **B. Bairwa**, P. Dharanidharan, and H. Rajini, "Designing and simulation for cooler section in cement industry," *IEEE NKCon*, 2022. doi:10.1109/NKCon56289.2022.10126915

2021

[C75] **B. Bairwa**, M. Sarvagya, S. Kumar, and A. Banik, "Battery swapping mobile service station for electric vehicles application," *2nd Int. Conf. on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, 2021. doi:10.1109/ICSTCEE54422.2021.9708560

[C76] **B. Bairwa**, S. Hampannavar, K. S. Vishal, and K. M. Bhargavi, "Modeling and simulation for capacity fade prediction of lithium-ion battery," *19th IEEE Int. Conf. on Smart Technologies (EUROCON)*, pp. 537–541, 2021. doi:10.1109/EUROCON52738.2021.9535594

[C77] S. Mansani, S. Hampannavar, K. N. Patil, K. V. L. Narayan, **B. Bairwa**, and E. Ayenew, "Wind speed assessment of three climate zones of Ethiopia," *3rd IEEE Int. Virtual Conf. on Innovations in Power and Advanced Computing Technologies (i-PACT)*, 2021. doi:10.1109/i-PACT52855.2021.9696756

2020

[C78] **B. Bairwa**, A. Soni, and K. Pareek, "Higher order equivalent circuit model analysis of lithium ion battery for electric vehicle," *AIP Conference Proceedings*, vol. 2294, 2020. doi:10.1063/5.0031346

[C79] **B. Bairwa**, K. Kumar, A. Soni, and K. Pareek, "MARS based state of charge estimation using real-life loading condition of lithium-ion battery for electric vehicle," *AIP Conference Proceedings*, vol. 2294, 2020. doi:10.1063/5.0031347

INVITED TALKS AND RESOURCE-PERSON ENGAGEMENTS

- **2026** – Expert Talk, National Institute of Technology Karnataka (NITK), Surathkal; session: *Battery Management Systems*, focusing on BMS fundamentals, battery monitoring and management, and emerging technologies in electric mobility.
- **2026** – Resource Person, National Institute of Technology Karnataka (NITK), Surathkal; session: *Electric Vehicle Modeling Using MATLAB and Simulink*, covering EV modeling fundamentals, driving scenario modeling, and hands-on exposure to RoadRunner and Virtual Vehicle Composer for simulation-driven development, 22 April 2026.
- **2026** – Resource Person, FDP/SDP on "Transformative Innovations in Chemistry through Multidisciplinary Collaboration"; session: *Battery Chemistry Fundamentals and MATLAB-Based Modeling*, School of Applied Sciences, REVA University, Bengaluru.
- **2026** – Resource Person, Faculty Development Session for the School of Agricultural Engineering, REVA University, Bengaluru; session: *Electric Vehicle Technologies and Motor Drive Control*, covering EV powertrain components, motor drive technologies, and MATLAB/Simulink-based modelling and simulation with applications in Agricultural Engineering, 1 July 2026.
- **2026** – Resource Person, One-Week FDP on "Innovative Research Prospects in Electrical Systems Integrating Artificial Intelligence (AI)"; session: *Dynamic Systems Modeling & Control: A Hands-On Approach Using MATLAB Simscape and AI Tools*, Dr. D. Y. Patil Institute of Technology, Pune.
- **2025** – Resource Person, MathWorks Usability Research Summit; session: *Hardware Integration to MATLAB*.
- **2025** – Resource Person, FDP Expert Session: *System Modelling Paradigms, Modelling Dynamic Systems, Physical System Modelling with Simscape and Control System Design*, Don Bosco Institute of Technology (DBIT), Bengaluru.
- **2025** – Expert Session: *Battery Management Systems Using AI and Machine Learning Techniques*, Electronics & ICT Academy (NIT Warangal in association with NIT Durgapur).
- **2025** – Resource Person, FDP: *Battery Pack Design and Simulation for Electric Vehicles – A Modeling Approach*, Bangalore Institute of Technology (BIT), Bengaluru.
- **2025** – Resource Person, FDP: *Battery Pack Design and Simulation for Electric Vehicles – A Modeling Approach*, Dayananda Sagar College of Engineering (DSCE), Bengaluru.
- **2025** – Expert Session: *Flight Dynamics and Control Simulation for Aerospace Applications Using MATLAB Simulink*, Aerospace Engineering, REVA University, Bengaluru.
- **2025** – Workshop: *MATLAB for Engineers: A Hands-on Approach to Modeling & Simulation*, School of EEE, REVA University, Bengaluru.
- **2024** – Expert Speaker, AICTE ATAL FDP: *Advanced Battery Modeling Techniques for Electric Vehicles with Emphasis on Renewable Energy Integration*, Holy Mary Institute of Technology and Science, Hyderabad.
- **2024** – Resource Person, AICTE ATAL FDP: *Mathematical Modeling of Lithium-Ion Battery for Electric Vehicle Application*, KITS Guntur, Andhra Pradesh.
- **2024** – Resource Person: *Battery Pack Design and Simulation for Electric Vehicles – A Modeling Approach*, Department of EEE, DSCE, Bengaluru.
- **2024** – Workshop Speaker: *Lithium-Ion Battery Modeling for Electric Vehicle Application*, Madanapalle Institute of Technology & Science, Andhra Pradesh.
- **2024** – Resource Person, FDP: *Battery Modeling and Parameter Estimation for Electric Vehicle Applications*, Guru Nanak Institutions Technical Campus, Hyderabad.
- **2024** – Keynote Speaker, FDP: *Parameter Estimation of Electric Vehicle Battery*, SJB Institute of Technology, Bengaluru.
- **2024** – Keynote Speaker, FDP: *Electric Vehicle Battery Modeling and Parameters Estimation*, Vidya Vikas Institute of Engineering and Technology, Mysuru.
- **2024** – Keynote Speaker, International Symposium and Workshop: *Sustainable Technology for a Sustainable Future 2024*, Modern

Education Society's College of Engineering (MESCOE), Pune.

- **2024** – Seminar Speaker: *Modeling of Electric Vehicles Motors*, School of Mechanical Engineering, REVA University, Bengaluru.
- **2023** – Keynote Speaker, AICTE ATAL FDP: *Parameter Estimation of Electric Vehicle Battery*, M. S. Ramaiah Institute of Technology, Bengaluru.
- **2023** – Keynote Speaker, International Symposium and Workshop: *Lithium-Ion Battery Modeling*, MESCOE, Pune.
- **2023** – Seminar Speaker: *Lithium-Ion Battery Modeling for Electric Vehicle Applications*, School of Mechanical Engineering, REVA University, Bengaluru.
- **2022** – Webinar Speaker: *Lithium-Ion Battery Modeling for Electric Vehicle Applications*, BMS College of Engineering, Bengaluru.
- **2022** – Webinar Speaker: *Lithium-Ion Battery Modeling for Electric Vehicle Applications*, Mewar University, Chittorgarh.

TEACHING EXPERIENCE

Courses Taught at REVA University

Semester	Course	Students
Even 2026	Robotics and Application	28
Odd 2025	Robotics and Application	70
Even 2025	Electric Vehicle System Engineering	60
Odd 2024	Energy Storage and Control Techniques	60
Even 2024	Electric Vehicle System Engineering	60
Odd 2023	Energy Conservation (Open Elective)	60
Even 2023	Energy Storage	60
Odd 2023	Green Building Technology	25
Even 2022	Fundamentals of Robotics	60
Odd 2022	Smart Grid and Applications	25
Even 2022	Python Programming	60
Odd 2021	Energy Storage and Control Techniques	25
Even 2021	Machine Learning with MATLAB	60

Laboratory Courses Handled

MATLAB and Simulink Lab (*Lab In-charge*) • Python Programming Lab (*Lab In-charge*) • Signal Processing Lab (*Lab In-charge*) • Microcontroller Lab • Digital System Design Lab • Basic Electrical and Electronics Engineering Lab

Certification Courses and Workshops Offered as Instructor

Year	Title	Duration	Participants
2026	Certification Course: Electric Vehicle Powertrain Modeling, Control Design, and Embedded Hardware Deployment Using MATLAB and Simulink	40 hours	-56
2026	Advanced LaTeX Workshop for Research Paper Writing	6 hours	25
2025	MATLAB for Engineers: A Hands-on Approach to Modeling & Simulation	2 days	35
2024	Mastering Simulink: From Basics to Advanced Modeling Techniques in MATLAB	32 hours	76
2024	Modeling and Simulation of Powertrain, Battery and Control System Using MATLAB (2-day workshop), MESCOE, Pune	16 hours	40

RESEARCH SUPERVISION

Undergraduate projects guided: 13 (2021–2025) | Postgraduate (M.Tech.) projects guided: 5 (2022–2025)

Year	UG	PG	Major Outcomes
2025	04	02	IEEE publications; multiple KSCST grants; prototypes developed; M.Tech. projects ongoing
2024	02	01	IEEE publication; KSCST grant; Project Expo winner; 2 Scopus-indexed publications (PG)
2023	03	01	IEEE/Scopus-indexed publications (including 2 papers from a single project); KSCST grant (PG)
2022	02	01	Scopus-indexed publications; KSCST project grant
2021	02	–	2 Scopus-indexed publications

ACADEMIC AND ADMINISTRATIVE RESPONSIBILITIES

KSCST Project Coordinator (School Level)

Year	Projects Sanctioned	Amount (INR)	Remark
2024–25	13	64,000	Highest in Karnataka
2023–24	14	71,000	Highest in Karnataka
2022–23	06	38,000	
2021–22	02	20,000	

Other Institutional Roles

- **GATE Coordinator (School of EEE):** guided 8 students to qualify GATE 2024; GATE 2025 qualifiers include All India Ranks 293 (IN) and 378 (EE).
- **Industry Collaboration Coordinator:** Bosch Global Software Technologies (2023); MathWorks/MATLAB (2025); CoreEL Technologies (2025).
- **Social Media Coordinator, School of EEE:** manages the School's LinkedIn, Instagram, Facebook and WhatsApp channels to promote academic events, achievements and industry engagement.
- **Coordinator roles:** Expert Talk, Seminar, Webinar, Workshop, FDP, Certification Course, Newsletter, Website and UPSC coordination; SDP Coordinator (until 2022).
- **Member**, AI First University team, REVA University.
- Professional network of **27,000+ followers** on LinkedIn, leveraged for internships, sponsored research and academia–industry collaboration.

CONFERENCES, FDPS AND SKILL DEVELOPMENT PROGRAMMES ORGANISED

Conferences

- **Editor**, International Conference on Recent Trends in Electrical, Electronics & Computer Engineering for Environmental and Sustainable Development (ICRTEEC-2021), *AIP Conference Proceedings*, vol. 2461, 2022.
- **Organising Committee Member**, IEEE International Conference on Trends in Electrical, Electronics, Computer Engineering (TEECCON) – 2022, 2023 and 2024.

Faculty Development Programmes (Coordinator)

- *Recent Trends in Electrical Power System*, REVA University – 2023 (45 participants) and 2024 (45 participants).

Skill Development Programmes (Coordinator)

Year	Programme	Days	Participants
2023	Power Electronics Converter for Industrial Application	4	240
2022	Machine Learning Using Python Programming	4	240
2021	Cyber Security and Python	4	240
2021	Solar, Battery Management and MEMS Technology	4	240

PROFESSIONAL SERVICE

- **Session Chair:** IEEE ICDCECE-2024, BITM Ballari (2024); IEEE International Conference AIKIIE-2023 (2023); IEEE ICAECC, REVA University, Bengaluru (2023).
- **Reviewer:** IEEE International Conference on Electrical, Computer and Energy Technologies, Sydney, Australia (2024); IEEE ICDCECE, BITM Ballari (2024, 2025); IEEE ICICACS, Raichur (2023); IEEE 3rd Global Conference for Advancement in Technology (GCAT), Bengaluru (2022).

TECHNICAL SKILLS

Modelling & Simulation	MATLAB scripting, Simulink block programming, Simscape and Simscape Battery, Simulink Design Optimization, MATLAB Coder, PVsyst
Programming & AI/ML	Python; deep learning for battery SoH (MLP, GRU, TCN, LSTM); parameter estimation by non-linear least squares; gradient-descent, active-set, interior-point and Levenberg–Marquardt optimisation; online feature extraction and classification
Embedded & IoT	Arduino, Raspberry Pi, NodeMCU, hardware integration with MATLAB, real-time control
Laboratory & Hardware	Battery cycling and analysis (Cadex battery analyser), electrochemical impedance spectroscopy (EIS) workstation, voltage/current sensor calibration, power-electronics hardware (DC–DC converters, inverters, gate-driver circuits)
Industrial Automation	PLC programming, HMI and SCADA
Documentation	LaTeX, OriginLab, Statistica, MS Visio, MS Excel, EndNote
Languages	Hindi and Rajasthani (native); English; Kannada (beginner)

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Year	Programme / Certification	Issuing Organisation
2026	Generative AI and LLMs: Architecture and Data Preparation	IBM
2026	Generative and Agentic AI with MATLAB and Simulink for Engineering	MathWorks
2026	Modernizing Engineering Education with MATLAB, Simulink and Generative AI	MathWorks
2026	Teaching with MATLAB	MathWorks
2025	MATLAB EXPO 2025 (Delegate, ID: 50377)	MathWorks
2025	Advanced Optimization Techniques using MATLAB (MNITJ/AOT/208/2025)	MNIT Jaipur
2025	MATLAB Coder Onramp; Introduction to Linear Algebra with MATLAB	MathWorks
2025	Generative AI for Beginners	Great Learning
2024	Battery Pack Modeling; Motor Control Modeling; Robotic Multibody Simulation; Simscape Battery Onramp; MATLAB Onramp	MathWorks
2024	Member of the Conference (MCA)	OPAL-RT Technologies
2021	FDP on Recent Trends in Smart Grid and Renewable Energy Technology	REVA University
2021	FDP on Challenges in Strengthening Clean Technology Entrepreneurship	MNIT Jaipur
2021	Certificate Program	Pandit Deendayal Energy University
2020	Conference on Flexible Electronics for Electric Vehicles (FlexEV-2020)	Manipal University Jaipur
2019	Energy Storage: The Future of Energy	AICTE ATAL Academy
2013	PLC Programming, HMI & SCADA	IDMI

PROFESSIONAL MEMBERSHIPS

- Member, Institute of Electrical and Electronics Engineers (**IEEE**), Membership No. 98476814 – since 2022
- Member, IEEE Power & Energy Society (**PES**) – since 2022
- Member, Solar Energy Society of India (**SESI**) – since 2022

REFERENCES

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DECLARATION

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Place: Bengaluru
Date: November 12, 2026

Dr. Bansilal Bairwa